

Problem 5141

Proposed by Kang Hyeonbin.

Let

$$f(x) = x(x+a)(x-a), \quad g(x) = x(x+2b)(x-2b),$$

where $a, b > 0$ are real constants. Suppose there exists a constant k such that the equation

$$|f(g(x))| = k$$

has exactly seven distinct real solutions. Find $\frac{b^3}{a}$.